project_description.md: Moroccan Adaptive AI Tutor Platform

1. Core Concepts & Features

Overview

The platform is an Adaptive AI guide (featuring an avatar) designed specifically for Moroccan students, covering all educational levels from Primary School through High School (2Bac). It operates seamlessly across Modern Standard Arabic (MSA) and French, depending on the subject matter.

Standard Student Tier

- **Curriculum Alignment:** The AI's knowledge base is strictly linked with the Moroccan core program descriptions, specifically the Pedagogy (البيداغوجية) and the Frame of Reference (الإطار المرجعي).
- **Pre-Loaded Knowledge:** Students can specify their educational level and specialization to access a full program of courses, exercises, exam examples, and a comprehensive exam archive from a proprietary dataset.
- **Custom Workspace:** Students can create custom fields to upload their own documents, courses, exercises, and PDFs.
- **Logic Resolution:** The system analyzes uploaded documents to provide solutions, step-by-step explanations, and generates similar exercises/exams for practice.

Private School & Educator Tier

- **Student Level Detection:** Upon registration, the system uses a diagnostic set of build-up questions to evaluate the student's baseline level of understanding.
- **Teacher Dashboard:** Educators can create virtual classrooms, invite students, and track their overall scores and developmental progress.
- **Adaptive Teaching:** Teachers use these analytics to adapt their teaching tools and techniques to the specific needs of each student in the classroom.

2. Planned Model Architecture

To handle complex logic resolution for math, physics, and science, alongside complex PDF analysis, the following AI stack will be utilized:

- **Data Extraction / PDF Parser:** Qwen2.5-VL-7B. This Vision-Language model will "see" and extract complex math equations, physics diagrams, and formatted text from raw PDFs, converting them into structured Markdown.
- **Core Reasoning Engine:** DeepSeek-R1-Distill-Qwen-14B. This logic and reasoning model will handle the heavy computational tasks (math, physics, and science resolution). It will

be fine-tuned using LoRA via the Unsloth library to understand the specific style and logic of the Moroccan curriculum.

- **Retrieval-Augmented Generation (RAG):** A vector database (ChromaDB or FAISS) will index course PDFs. This ensures the Reasoning Engine retrieves the exact Moroccan course context before answering, preventing hallucinations regarding local content.
- **Fast Inference/Routing (Optional):** Lighter Hugging Face Inference API models (e.g., Mistral/Qwen variants) for simple conversational routing or processing straightforward text queries before passing complex math to the 14B model.

3. Data Strategy

A. Datasets to Build from Scratch (Proprietary Asset)

Building the custom Moroccan curriculum dataset is the most valuable asset of this project. This requires scraping, cleaning, and formatting data into a strict (question, reasoning steps, answer) structure.

- **The Core Curriculum:** Courses, exercises, and exam examples for all subjects from Primary to High School.
- **The Guidelines:** Moroccan pedagogy rules and the official Frame of Reference (الإطار المرجعي) for all educational levels.
- **The Exam Archive:** National exam PDFs (Math, Physics, Sciences in French; Literature, History, Philosophy, Islamic Education in MSA).
- **Resources for Building:** National education portals and existing school resources will be scraped. Data will be processed on Kaggle or Google Colab using Qwen2.5-VL-7B to extract structured text, which will then be pushed directly to a private Hugging Face Dataset repository.

B. Ready-to-Use Datasets (For Pre-training & General Knowledge)

While your proprietary dataset handles Moroccan specifics, the models will rely on existing pre-trained knowledge for general logic, translation, and text generation.

- **General Math/Logic Datasets:** Open-source datasets on Hugging Face (e.g., *OpenMathInstruct*, *NuminaMath*) can be used if baseline math logic needs reinforcement.
- **Language Models Data:** Existing corpora for French, English, and Modern Standard Arabic (MSA) form the baseline of the models selected.

4. General MVP Build Plan (Standard Roadmap)

This roadmap outlines a standard, scalable development cycle to build out the full MVP architecture logically.

Phase 1: Data Engineering & Pipeline Setup

- **Objective:** Automate the conversion of raw curriculum PDFs into a usable machine-learning format.
- **Actions:** * Set up a Hugging Face organization/account to host private datasets and models.
  - Deploy Qwen2.5-VL-7B on Kaggle (utilizing free GPU hours).
  - Write Python scripts to parse the uploaded Moroccan exam/course PDFs and output clean, chunked Markdown.
  - Establish a pipeline that automatically pushes this structured data to your Hugging Face dataset.

Phase 2: RAG Integration & Knowledge Base Development

- **Objective:** Build the retrieval system to ground the AI in the Moroccan curriculum.
- **Actions:**
  - Initialize a Vector Database (ChromaDB or FAISS).
  - Create an embedding pipeline that takes the structured course chunks from Phase 1 and indexes them into the database.
  - Develop the search function: When a query is made, the system must accurately retrieve the relevant lesson or Frame of Reference snippet.

Phase 3: Model Fine-Tuning

- **Objective:** Teach the reasoning model the specific QA logic expected in Moroccan schools.
- **Actions:**
  - Deploy a cloud GPU instance on RunPod (RTX 3090/4090) for cost-effective heavy lifting.
  - Use the Unsloth library to efficiently fine-tune DeepSeek-R1-Distill-Qwen-14B using LoRA on the Question-Answer-Reasoning pairs generated in Phase 1.
  - Export and host the fine-tuned model weights on Hugging Face.

Phase 4: Platform & UI Development

- **Objective:** Build the user-facing application for both Standard and Private School tiers.
- **Actions:**
  - Develop the frontend application.
  - Build the user authentication and onboarding flow (level/specialization selection).
  - Implement the diagnostic build-up questionnaire for level detection upon registration.
  - Build the main student dashboard (course viewing, document upload module, AI chat interface).

- Build the teacher dashboard (classroom creation, student invites, progress tracking analytics).

Phase 5: System Integration & Testing

- **Objective:** Connect the frontend application to the AI infrastructure.
- **Actions:**
  - Host the fine-tuned Reasoning Engine via an API endpoint (e.g., RunPod Serverless or dedicated pod).
  - Wire the frontend chat/upload interface to the RAG pipeline, ensuring queries first fetch context from the Vector DB before being sent to the LLM endpoint.
  - Test the logic resolution with complex math/physics PDFs across French and MSA to ensure accurate extraction and reasoning.